ST512-SP26-Homework-5

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Problem 1

a

Source	df	SS	MS	F-value	p-value
Factor A	4	375	93.75	27.79	3.66e-14
Factor B	2	462	231.0	68.49	0
AxB Interaction	8	38	4.75	1.40	0.210
Error	75	253	3.373	–	–
Total	89	1128	–	–	–

b

Based on the ANOVA table, the interaction factor has a p-value of 0.210. Using a 0.05 significance level, we fail to reject the null hypothesis. Therefore, there is NOT a significant interaction between factor A and factor B.

c

Based on the ANOVA table, for Factor A the p-value was calculated to be 3.66e-14. Using a significance value of 0.05, we reject the null hypothesis. Therefore, there is a significant difference in mean response across the levels of factor A.

d

Based on the ANOVA table, for Factor B the p-value was calculated to be 0. Using a significance value of 0.05, we reject the null hypothesis. Therefore, there is a significant difference in mean response across the levels of factor B.

Problem 2

a

a=4, b=3, n=3, N=36, SSE=720

df (error)= 24, MSE=720/24=30.0

No interaction –> each factor A mean averages over b*n=9

Tukey MSD:

$q_{\alpha,4,24} * \sqrt{MSE/n_A)} = 3.90 * \sqrt{30/9} = 7.12$

Bonferroni MSD:

$t_{\frac{\alpha}{2C},24} * \sqrt{2MSE/n_A} = 2.875 * \sqrt{60/9} = 7.42$

b

Interaction –> compare Factor A within one level of Factor B, n=3

Tukey MSD:

$q_{\alpha,4,24} * \sqrt{MSE/n_A)} = 3.90 * \sqrt{30/3} = 12.33$

Bonferroni MSD:

$t_{\frac{\alpha}{2C},24} * \sqrt{2MSE/n_A} = 2.875 * \sqrt{60/3} = 12.86$

Problem 3

a

$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$

i=1,2,3 (variety); j=1,2,3(temp); k=1,…,8

Y_{ijk}= total dry weight of plant material

μ = overall mean

α_i = effect of variety i, $\sum \alpha_i = 0$

β_j= effect temperature j, $\sum \beta_j = 0$

$(\alpha\beta)_{ij}$ = interaction effect subject to $\sum(\alpha\beta)_{ij} = 0$

ϵ_{ijk} = random error ~N(0, σ^2)

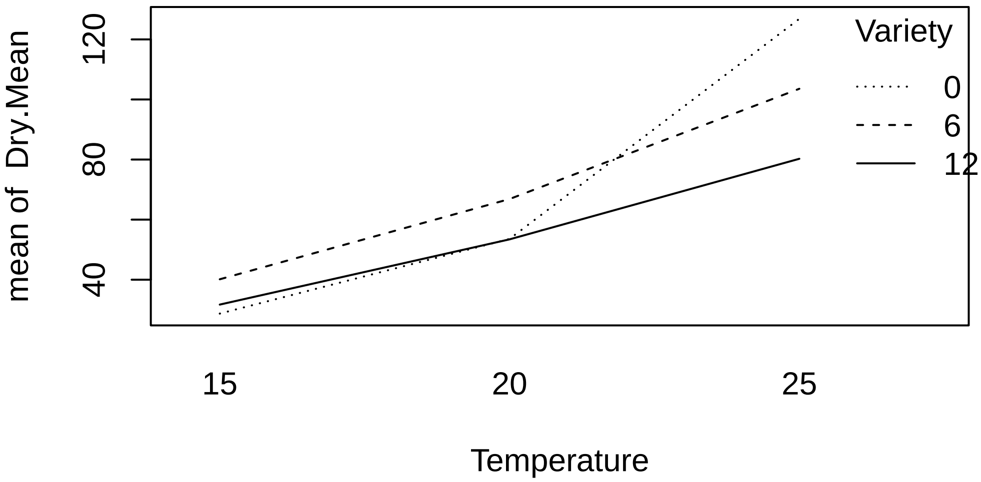
assumptions: independence, normality, equal variance

b

Source	df	SS	MS	F-value	p-value
model	8	71692.11	8961.51	189.01	<0.0001
Variety	2	3325.78	1662.89	35.07	<0.0001
Temperature	2	61549.77	30774.86	649.08	<0.0001
Var x Temp	4	6817.6	1704.4	35.95	<0.0001
Error	63	2987.02	47.413	–	–
Total	79	146372.28	–	–	–

c

```
[1] "Variety"     "Temperature" "Dry.Mean"
```



All three varieties increase with temperature, but at different rates. The lines cross, indicating a significant interaction. Variety 0 is lowest at 15°C but highest at 25°C.

d

$H_o : all(\alpha\beta)_{ij} = 0$

$H_A : atleastone(\alpha\beta)_{ij} \neq 0$

F-value: $\frac{MS_{AB}}{MSE}$ = 1704.4/47.413= 35.95

1-pf(35.94,4,63) = p value = 1.33e-15

There is a significant interaction between Variety and Temperature.

Problem 4

Source	df	SS	MS	F-value	p-value
Climate	2	1255.24	627.62	11.13	0.0002
Location	2	98.63	49.315	0.875	0.4255
Interaction	4	882.76	220.69	3.95	0.0092
Error	36	2029.68	56.38	–	–
Total	44	4266.31	–	–	–

$H_o : \frac{\mu_{11}+\mu_{12}}{2} = \frac{\mu_{31}+\mu_{32}}{2}$

$L = 0.5\mu_{11} + 0.5\mu_{12} + (-0.5)\mu_{31} + (-0.5)\mu_{32} = 0$

$\hat{L} = 0.5\bar{y}_{11.} + 0.5\bar{y}_{12.} + (-0.5)\bar{y}_{31.} + 0.5\bar{y}_{32.}$

$\hat{L}$ = 0.5(22.8) + 0.5(19.8) + (-0.5)(28) + (-0.5)(26) = -5.7

V($\hat{L}$) = MSE $\sum \frac{c_i^2}{n}$ = 56.38 * 4/5 = 45.10

SE($\hat{L}$) = 6.72

t = $\frac{\hat{L}}{SE(\hat{L})}$ = -5.7/6.72 = -0.848

p-value = 0.201 > 0.05; FTR null

There is not a significant difference between the number of bites in the arm or neck in cold and and humid climates.

Problem 5

N=299, df(error)= 295, MSE= $\frac{42088}{295}$= 142.67

a

Comparing model 1 and 2.

F = $\frac{42122-42088}{142.67}$ = 0.238

p.value= 1-pf(0.238,1,295) = 0.626

There is NOT a significant interaction between Gender and Rank.

b

Comparing model 1 and 3.

F = $\frac{785}{142.67}$ = 5.50

p.value= 1-pf(5.50,1,295) = 0.0197

There IS a significant main effect of Gender.

R code

problem 1

```r
1-pf(27.79,4,75)
```

```
[1] 3.663736e-14
```

```r
1-pf(68.49, 2,75)
```

```
[1] 0
```

```r
1-pf(1.40,8,75)
```

```
[1] 0.2105261
```

problem 2

```r
qtukey(.95,4,24)
```

```
[1] 3.901262
```

```r
qt(0.0041667,24)
```

```
[1] -2.875091
```

problem 3

```r
1-pf(35.95,4,63)
```

```
[1] 1.332268e-15
```

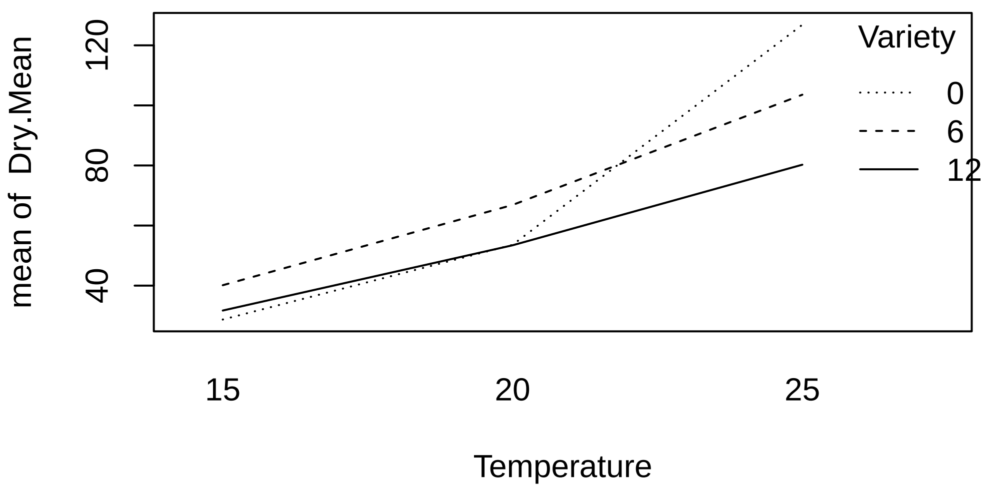
```
plant_data<-read.csv("plant_data.csv",header=TRUE)
attach(plant_data)
```

```
The following objects are masked _by_ .GlobalEnv:

    Temperature, Variety

The following objects are masked from plant_data (pos = 3):

    Dry.Mean, Temperature, Variety
```

```
names(plant_data)
```

```
[1] "Variety"     "Temperature" "Dry.Mean"
```

```
Variety<-factor(Variety)
Temperature<-factor(Temperature)
interaction.plot(Temperature, Variety, Dry.Mean)
```



problem 4

```
1-pf(11.13, 2, 36)
```

```
[1] 0.0001724949
```

```r
1-pf(0.875,2,36)
```

```
[1] 0.4255394
```

```r
1-pf(3.95,4,36)
```

```
[1] 0.009284901
```